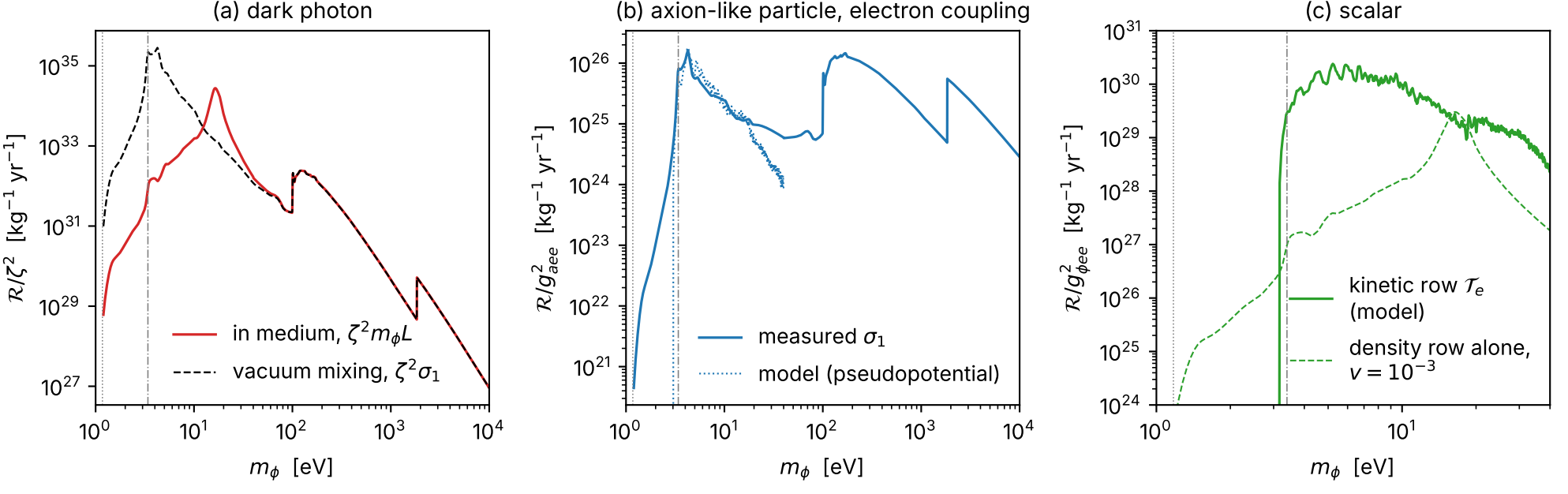
(a) dark photon
$\mathcal{R}/\zeta^2$ [kg^{-1} yr^{-1}]
10^{35}
10^{33}
10^{31}
10^{29}
10^{27}
in medium, $\zeta^2 m_\phi L$
vacuum mixing, $\zeta^2 \sigma_1$
10^0
10^1
10^2
10^3
10^4
m_ϕ [eV]
(b) axion-like particle, electron coupling
$\mathcal{R}/g_{aee}^2$ [kg^{-1} yr^{-1}]
10^{26}
10^{25}
10^{24}
10^{23}
10^{22}
10^{21}
measured σ_1
model (pseudopotential)
10^0
10^1
10^2
10^3
10^4
m_ϕ [eV]
(c) scalar
$\mathcal{R}/g_{\phi ee}^2$ [kg^{-1} yr^{-1}]
10^{31}
10^{30}
10^{29}
10^{28}
10^{27}
10^{26}
10^{25}
10^{24}
kinetic row $\mathcal{T}_e$ (model)
density row alone, $v = 10^{-3}$
10^0
10^1
m_ϕ [eV]